# Description

## Illustration





## Figure caption

1. Asphalt concrete freshly laid: the aggregate is visible. 2. Cracking damage to an asphalt concrete road surface shows the material within. Images used under license from Shutterstock.com

## The material

Asphalt concrete is a mix of bitumen and aggregate. A macadam-type surfacing, it is used for paths, pavements, roads, and runways. The mechanical properties vary strongly with temperature, time of loading and with the grade of bitumen. The data below are for 20 C, the moduli are for dynamic loading and the strengths are for a loading time of about 1000 seconds. All must be regarded as approximate. Mix constituents and proportions are varied to produce widely ranging properties. It is often simply called 'asphalt' and sometimes referred to as Tarmac or Tarmacadam for historical reasons, though does not contain tar.

## Compositional summary

A mix of sand and aggregate with bitumen (asphalt). Typical bitumen content: 4-10% by weight. Rubbery polymers are sometimes added to the mix to improve properties.

# General properties

| | | | | | |
|---|---|---|---|---|---|
| Density | | 2.2e3 | - | 2.8e3 | kg/m^3 |
| Price | * | 0.0151 | - | 0.0604 | GBP/kg |
| Date first used | | -625 | | | |

# Material form that data applies to

| | |
|---|---|
| Bulk | ✓ |

# Building system

| | |
|---|---|
| Superstructure | ✓ |
| Enclosure | ✓ |
| Interiors | ✗ |

| | | | | |
|---|---|---|---|---|
| Services | ✖ | | | |

## Mechanical properties

| | | | | |
|---|---|---|---|---|
| Young's modulus | 10 | - | 50 | GPa |
| Shear modulus | * 3 | - | 15 | GPa |
| Bulk modulus | * 10 | - | 50 | GPa |
| Bending modulus | * 8 | - | 40 | GPa |
| Poisson's ratio | * 0.35 | - | 0.36 | |
| Yield strength (elastic limit) | * 0.1 | - | 0.3 | MPa |
| Tensile strength | 0.1 | - | 0.3 | MPa |
| Compressive strength | * 0.5 | - | 1.3 | MPa |
| Bending strength | * 0.1 | - | 0.3 | MPa |
| Elongation | 0.4 | - | 2 | % strain |
| Hardness - Vickers | * 0.4 | - | 0.6 | HV |
| Fatigue strength at 10^7 cycles | * 0.04 | - | 0.12 | MPa |
| Fracture toughness | 0.3 | - | 3 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | * 0.3 | - | 0.8 | |

## Thermal and combustion properties

| | | | | |
|---|---|---|---|---|
| Thermal conductor or insulator? | Poor insulator | | | |
| Thermal resistivity | * 0.5 | - | 0.7 | m.°C/W |
| Thermal expansion coefficient | * 12 | - | 14 | µstrain/°C |
| Specific heat capacity | * 850 | - | 920 | J/kg.°C |
| Melting point | * 9.85 | - | 49.9 | °C |
| Maximum service temperature | * 49.9 | - | 125 | °C |
| Minimum service temperature | * -60.2 | - | -0.15 | °C |
| Flammability | Self-extinguishing | | | |
| Emissivity | 0.9 | | | |

## Hygro-thermal properties

| | | | | |
|---|---|---|---|---|
| Water absorption | 0 | | | % |
| Water vapor permeability | 0 | | | kg.m/s.m^2.Pa |
| Frost resistance | Average | | | |

## Electrical properties

| | | | | |
|---|---|---|---|---|
| Electrical conductor or insulator? | Good insulator | | | |
| Electrical resistivity | 1e18 | - | 1e20 | µohm.cm |
| Dielectric constant (relative permittivity) | 2.7 | - | 3 | |
| Dissipation factor (dielectric loss tangent) | 0.005 | - | 0.05 | |

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

| | | | | |
|---|---|---|---|---|
| Dielectric strength (dielectric breakdown) | 10 | - | 20 | MV/m |

## Optical properties

### Natural color

Black or gray

| | | | | |
|---|---|---|---|---|
| Transparency | Opaque | | | |
| Transmissivity | 0 | | | % |

## Critical Materials Risk

| | |
|---|---|
| High critical material risk? | No |

## Acoustic properties

| | |
|---|---|
| Sound absorption | Good |
| Sound isolation | Good |

## Processability (scale 1 = impractical to 5 = excellent)

| | | | | |
|---|---|---|---|---|
| Castability | 3 | - | 4 | |
| Moldability | 3 | - | 4 | |

## Durability

| | |
|---|---|
| Water (fresh) | Excellent |
| Water (salt) | Excellent |
| Weak acids | Excellent |
| Strong acids | Limited use |
| Weak alkalis | Excellent |
| Strong alkalis | Acceptable |
| Organic solvents | Unacceptable |
| UV radiation (sunlight) | Excellent |
| Industrial atmosphere | Excellent |
| Rural atmosphere | Excellent |
| Marine atmosphere | Excellent |

## Primary material production: energy and climate change

| | | | | | |
|---|---|---|---|---|---|
| Climate change ($CO_2$-eq) (virgin grade production) | * 0.0824 | - | 0.0909 | kg/kg |
| Embodied energy (virgin grade production) | * 3.92 | - | 4.33 | MJ/kg |
| Water Usage | * 3.62 | - | 4 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Machining, grinding, embodied energy | * 14.7 | - | 16.2 | MJ/kg |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Machining, grinding, climate change (CO2-eq) | * 0.959 | - | 1.06 | kg/kg |

## Recycling and end of life

| | | | | |
|---|---|---|---|---|
| Recycle | ✓ | | | |
| Functional recycle | ✗ | | | |
| Climate change (CO2-eq), recycling | * 0.00824 | - | 0.00909 | kg/kg |
| Embodied energy, recycling | * 0.549 | - | 0.606 | MJ/kg |
| Recycle fraction in current supply | 14 | | | % |
| Combust for energy recovery | ✗ | | | |
| Landfill | ✓ | | | |
| Biodegrade | ✗ | | | |
| A renewable resource? | ✗ | | | |

## Supporting information

### Design guidelines

The properties of bituminous mixes depend on the volume fraction, the shape and size grading of aggregate and the grade of bitumen. They are designed to resist permanent deformation, fatigue cracking, workability, and to provide a skid resistant surface. Bitumen, and the fumes it emits when heated, are toxic.

### Technical notes

Typical compositions, by volume %, of asphalt concrete are:

(Component; Coated stone; Continuously graded; Gap graded; Mastic)
Coarse aggregate; 64; 44; 26; 27
Fine aggregate; 5; 32; 46; 18
Filler; 2; 4; 8; 27
Bitumen; 8; 11; 17; 26
Void content; 20; 8; 3; <1

A high percentage of fine aggregate gives a smooth surface but poor skid resistance. To increase this, a greater proportion of coarse chippings is introduced.

### Typical uses

Road beds and pavements, runways.

### Tradenames

Asphalt, Tarmac

### Related standards

The properties of Cement and Concrete vary a great deal based on composition, time and processing. Standards exist to make sure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main EU and US standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

ASTM C1097: Standard Specification for Hydrated Lime for Use in Asphalt Cement or Bituminous Paving Mixtures
ASTM D1073: Standard Specification for Fine Aggregate for Asphalt Paving Mixtures
ASTM D2397: Standard Specification for Cationic Emulsified Asphalt
ASTM D3141: Standard Specification for Asphalt for Undersealing Portland-Cement Concrete Pavements
ASTM D3381: Standard Specification for Viscosity-Graded Asphalt Cement for Use in Pavement Construction
ASTM D4215: Standard Specification for Cold-Mixed, Cold-Laid Bituminous Paving Mixtures
ASTM D517: Standard Specification for Asphalt Plank
ASTM D6154: Standard Specification for Chemically Modified Asphalt Cement for Use in Pavement Construction
ASTM D6155: Standard Specification for Nontraditional Coarse Aggregates for Bituminous Paving Mixtures
ASTM D6373: Standard Specification for Performance Graded Asphalt Binder
ASTM D692: Standard Specification for Coarse Aggregate for Bituminous Paving Mixtures
ASTM D946: Standard Specification for Penetration-Graded Asphalt Binder for Use in Pavement Construction
ASTM D977: Standard Specification for Emulsified Asphalt
BS 594987: Asphalt for roads and other paved areas. Specification for transport, laying, compaction and product type testing protocols
EN 13108: Bituminous mixtures. Material specifications.

## Links

Producers

Reference